Privacy-Preserving Recommendations With Mixture Model-Based Matrix Factorization Under Local Differential Privacy

Pengfei Zhang, Hong Sun, Zhikun Zhang, Xiang Cheng, Youwen Zhu, and Ji Zhang

***Abstract*—Matrix factorization-based recommendations have emerged as a cornerstone for many industrial recommendation algorithms. However, studies employing differential privacy often rely on a trusted server, while those implementing local differential privacy (LDP) frequently encounter substantial accuracy degradation and excessive communication overhead due to noise injection and frequent user interactions. Moreover, the presence of injected noise for privacy protection and inherent Gaussian noise within these perturbed values compounds the accuracy issues and may create a cascading effect under LDP, exacerbating the complexity of the problem at hand. To address these challenges, we propose *MENTOR*, which is *M*ixture model-based r*E*comme*N*dations approach with matrix Fac*TOR*ization under LDP. Its main idea lies in the adoption of a bounded input perturbation mechanism that closely approximates the Laplace distribution while providing rigorous LDP, coupled with accounting for various noise types present in the perturbed data. This approach allows us to reformulate the matrix factorization process with minimal user interaction, requiring only a single round of communication. In particular, to add noise, we design a bidirectional bounded LDP input perturbation mechanism *BBV* while minimizing variance. To generate recommendation results, we devise a matrix factorization technique *GLMF* based on a Gaussian–Laplacian mixture model. Comprehensive experiments reveal that *MENTOR* outperforms the state of the art by at least 15% in *RMSE* and 12% in *F-Measure*, showcasing its effectiveness in balancing privacy and utility in recommendation systems.**

***Index Terms*—Local differential privacy (LDP), matrix factorization (MF), recommender system.**

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List of Frequently Used Notations

Notation	Definition.
ε	Privacy budget.
M	Total number of users.
N	Total number of items.
U and V	User and item latent space.
R and R^*	Private and nonprivate rating matrix.
I	Indicator for whether a worker has done a task.
K	Number of mixed distributions.
p	Probability for remaining real rating.
p	Real rating range to remain unchanged.
Ω	Observed rating space.
π_k	Weight of the kth mixed distribution.
σ_k	Standard deviation for kth mixed distribution.
γ_{ijk}	Indicator for whether R_ij belongs to the π_k.
α	Proportion of Gaussian noise.
λ	Regularization parameter.

I. Introduction

THE explosive growth of industrial e-commerce platforms has fundamentally transformed how products and services are recommended. As the industrial ecosystem becomes increasingly digitalized, users face an overwhelming array of choices across manufacturing services, industrial products [1]. This challenge has made recommendation systems an indispensable component of modern industrial platforms. Matrix factorization (*MF*) [2] has emerged as a cornerstone technique in industrial recommendation systems, decomposing complex user–item interactions into interpretable latent factors that capture both user preferences and item characteristics. However, implementing effective *MF* could lead to the exposure of sensitive industrial data that might lead to serious consequences, including competitive disadvantages, and compromised business strategies [3]. Thus, protecting user privacy while maintaining effective recommendations becomes an urgent need.

Local differential privacy (LDP) [4] has garnered significant attention in privacy-preserving recommendations, particularly in scenarios where servers cannot be fully trusted. This approach offers enhanced privacy safeguards compared to conventional differential privacy (DP)-based recommendation systems [5], [6], [7], [8], [9], [10], [11]. Notably, even in environments with trusted servers, the risk of privacy compromises during data transmission remains a significant concern [12]. To mitigate

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these vulnerabilities, a growing body of research [5], [12], [13], [14] has focused on achieving LDP in recommendation systems. In these approaches, users apply noise to their data locally before transmission, thereby preserving data integrity at its source.

Motivation: Despite the growing interest in LDP for recommendations, existing solutions [12], [15], [16], [17] face several critical limitations. Current privacy-preserving recommendation systems using LDP [12], [15], [16], [17] face several substantial limitations that affect their practical implementation and effectiveness.

1) *Privacy–utility tradeoff:* Traditional approaches, such as the Laplace mechanism [15], [16], use unbounded distributions for noise injection, which creates two major problems. First, these methods can generate extreme outlier values that fall outside valid rating ranges (such as 1–5 stars), producing meaningless results [12]. Second, they fail to consider the inherent patterns in user rating behavior [17], leading to unrealistic data perturbations that compromise recommendation quality even with generous privacy budgets [15].
2) *Communication overhead:* The decentralized nature of LDP systems [16], [17] creates significant implementation barriers. These systems require frequent user–server communications and extensive local computations on resource-constrained user devices [12], [15]. The need for multiple communication rounds to achieve convergence further compounds these issues [16], increasing both computational costs and potential privacy vulnerabilities. Each additional round of communication not only strains system resources but also provides more opportunities for potential privacy leakage [17].
3) *Complex noise interactions:* A fundamental challenge involves the interaction between two distinct types of noise: the intentionally added Laplacian noise for privacy protection [15], [16] and the inherent Gaussian noise that naturally exists in user preference data [18]. Current methods treat these noise sources independently [12], [17], failing to account for their compounding effects. This oversight leads to significant deterioration in recommendation quality as the combined noise makes it increasingly difficult to extract meaningful patterns from the data [15], [18].

Prior work most closely related to ours [15] attempts to address some of these issues but introduces new limitations. Their noise-addition approach emphasizes boundedness at the expense of noise minimization, which results in excessive noise injection [15], [16]. Furthermore, their use of Gaussian mixture models (GMMs) fails to account for the Laplacian characteristics of privacy-preserving noise [15], necessitating additional training iterations to achieve comparable accuracy levels. These limitations demonstrate the clear need for new approaches that can simultaneously provide strong privacy guarantees and maintain high recommendation accuracy while addressing practical implementation challenges [17], [18].

Our proposal: To address these fundamental challenges, we present Mixture model-based rEcommeNdations approach with matrix facTORization under LDP (*MENTOR*), a comprehensive solution that systematically tackles each limitation of existing approaches.

1) To tackle the privacy–utility tradeoff, *MENTOR* introduces *BBV*, a bidirectional bounded perturbation mechanism that operates within predefined limits while ensuring strict LDP guarantees. We frame this as a constrained optimization problem, deriving a closed-form solution that minimizes noise variance, effectively addressing unbounded perturbations and extreme outliers.
2) For system efficiency, *MENTOR* employs a single-round communication design, requiring users to transmit each perturbed rating only once—matching the efficiency of nonprivate systems [19] while reducing computational and privacy risks.
3) To enhance robustness against complex noise interactions, we propose *GLMF*, a Gaussian–Laplacian mixture model-based MF approach. By explicitly modeling both privacy-preserving and inherent Gaussian noise, and integrating them via expectation maximization (EM), *GLMF* significantly improves recommendation accuracy.
4) Unlike prior methods [15] that address either boundedness or Gaussian modeling in isolation, *MENTOR* unifies both, ensuring strong privacy with superior accuracy. In addition, its single-round design offers practical advantages over iterative alternatives.

Beyond recommendation systems, our work advances privacy-preserving machine learning. *BBV* introduces a novel framework for bounded perturbation applicable to any LDP-constrained optimization, while our variance minimization strategy establishes new foundations for handling bounded numerical data. Furthermore, *GLMF*'s mixed noise modeling provides a generalizable solution for systems affected by both natural and privacy-induced noise. These innovations extend to federated learning, distributed optimization, and privacy-preserving analytics.

Contributions: The key contributions of this work are as follows.

1) We introduce *MENTOR*, a novel framework that simultaneously tackles privacy–utility tradeoffs, communication overhead, and complex noise interactions in LDP-based systems. By integrating bounded perturbation with mixed noise modeling, *MENTOR* sets a new standard for privacy-preserving recommendations, extendable to online platforms.
2) We develop *BBV*, a constrained optimization-based perturbation mechanism that minimizes noise variance while ensuring strict LDP guarantees. Unlike traditional unbounded noise methods, *BBV* effectively mitigates extreme outliers, preserving both data privacy and utility.
3) We propose *GLMF*, a noise-aware recommendation model that jointly accounts for privacy-induced Laplacian noise and inherent Gaussian noise in user preferences. By leveraging EM, *GLMF* surpasses single-distribution approaches, enhancing both accuracy and robustness.

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4) Our theoretical analysis establishes *MENTOR*'s privacy guarantees, utility bounds, and computational efficiency. Empirical results on two real-world datasets show that *MENTOR* outperforms state-of-the-art methods by over 15% in *RMSE* and 12% in *F-Measure*, while maintaining the communication efficiency of nonprivate systems, validating its practical effectiveness.

The rest of this article is organized as follows. Section II summarizes the related work in privacy-preserving recommendation systems. Section III presents the problem formulation. Section IV presents our proposed *MENTOR* framework in detail, and Section V provides theoretical analysis of its privacy, utility, and complexity. Section VI reports comprehensive experimental results that demonstrate *MENTOR*'s superior performance compared to state-of-the-art methods. Finally, Section VII concludes this article and outlines future research directions.

II. Related Work

Early efforts estimate item similarity on the server [20], enhance accuracy via Condensed LDP [13], and leverage clustering in federated learning for adaptive training [12]. To mitigate noise, Zheng et al. [21] applied global rating averaging, while Wang et al. [16] used *Piecewise* [22] to protect both rating values and item sets, although at the cost of stability and high communication overhead. Other methods include lightweight shard gradient training [23], Gaussian mixture modeling under LDP [15], and graph convolutional networks for personalized LDP updates [17].

To improve accuracy with side information, researchers explore community-based attribute inference [4], privacy-preserving modeling of ratings and social ties [24], and protection of social relationships using location data [25]. Trust-based similarity [26] and privacy-aware POI recommendations [27] further highlight the potential of auxiliary data. However, integrating such information while ensuring strong privacy guarantees remains an open challenge.

In summary, existing LDP-based recommendation approaches [12], [15], [16], [17] face several limitations that hinder practical deployment. They struggle to balance prediction accuracy with strong privacy guarantees due to overlooking domain characteristics and inherent noise in data. Their decentralized nature leads to high communication overhead. Traditional mechanisms, such as Laplace noise, with unbounded distributions, can produce extreme, meaningless results even with generous privacy budgets. Moreover, they fail to address the compounded effect of multiple noise types, the injected Laplacian noise for LDP and inherent Gaussian noise from user preferences, which distorts recommendations. Systems requiring multiple rounds of data exchange also increase privacy risks and computational costs. The work closest to ours [15] focuses on bounded noise but suffers from large noise injection and inadequate modeling of Laplace noise, requiring more training. In contrast, we propose *MENTOR*, which uses a bounded input perturbation mechanism and models mixed noise distributions with a Gaussian–Laplacian mixture model, achieving high accuracy, strong LDP guarantees, and only a single round of communication.

III. Problem Statement

Consider a recommendation system with a set of M users $\mathcal{U} = u_1, \ldots, u_M$ and a set of N items $\mathcal{I} = i_1, \ldots, i_N$. Each item rating belongs to a bounded domain $D = [d_1, d_2]$, where d_1 and d_2 represent the minimum and maximum possible rating values (e.g., 1 and 5 in a five-star rating system), respectively. The system input consists of a sparse rating matrix $R \in \mathbb{R}^{M \times N}$, where each entry r_{ij} represents user i's rating for item j. Let $\Omega = (i, j)|r_{ij}$ is observed, denoting the set of observed user–item rating pairs.

For each user i, we define $\mathcal{R}_i = r_{ij}|(i, j) \in \Omega$ as their set of provided ratings. These ratings are considered private information that can reveal user preferences and behavioral patterns. To protect user privacy, we employ LDP with a uniform privacy budget $\varepsilon > 0$ across all users.

The research objective is to design an algorithm $\mathcal{M}$ that satisfies ε-LDP for each user's ratings and simultaneously minimizes the prediction error on unobserved ratings. Formally, for any predicted rating $\hat{r}_{ij}$, we aim to minimize the expected error $\mathbb{E}[|\hat{r}_{ij} - r_{ij}|]$ subject to the LDP constraint: $\forall i \in [1, M] \forall r_i, r_i' \in D^{|\mathcal{R}i|} \forall S \subseteq \text{Range}(\mathcal{M})$:$\Pr[\mathcal{M}(r_i) \in S] \leq e^{\varepsilon} \cdot \Pr[\mathcal{M}(r_i') \in S]$, where $\text{Range}(\mathcal{M})$ represents all possible outputs of algorithm $\mathcal{M}$, and S is any subset of these outputs. This constraint ensures that for any two possible rating vectors r_i and r_i' of user i, the probability of algorithm $\mathcal{M}$ producing an output in set S differs by at most a factor of e^{ε}.

The Nomenclature summarizes the notations used in this article.

IV. Our Proposal

A. Overview of MENTOR

To address the challenges of accuracy degradation and excessive communication overhead in privacy-preserving recommendation systems, we introduce *MENTOR*, a novel approach that integrates a bounded input perturbation mechanism with a noise-aware MF process, while requiring only a single interaction with users. At the core of *MENTOR* are two key components: *BBV*, a sophisticated local input perturbation mechanism, and *GLMF*, an advanced algorithm for noise-aware rating prediction. It offers significant advantages over existing solutions, notably enhancing predictive accuracy while maintaining strong user privacy guarantees. Moreover, *MENTOR*'s ability to generate personalized recommendations based on a single-user interaction substantially reduces communication overhead. Fig. 1 illustrates the overall structure of *MENTOR*.

B. Bidirectional Bounded Input Perturbation

To provide bounded noise, we design a bidirectional bounded input perturbation mechanism, *BBV*, which minimizes variance. This method ensures privacy protection while minimizing the impact on data utility by adding noise in a controlled manner

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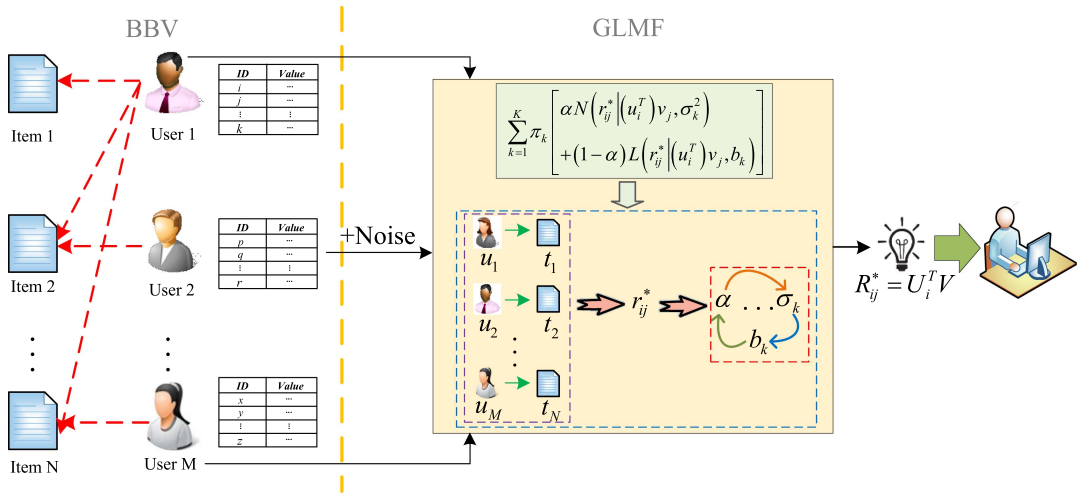


Fig. 1. Overview of *MENTOR*.

within predefined upper and lower bounds based on the specified domain.

Specifically, given t as the true value, which is the perturbation center, satisfying $d_1 \leq t-d$ and $t+d \leq d_2$, where $d>0$, the probability density function (pdf) of the perturbed value t' is shown as follows:

$$pdf\left(t'=x|t\right)=\begin{cases} p, & if x\in\left[\left(t-d\right),\left(t+d\right)\right] \\ \frac{p}{e^{\varepsilon}}, & if x\in\begin{bmatrix} d_1,\left(t-d\right))\cup \\ \left(\left(t+d\right),d_2\right] \end{bmatrix} \end{cases} \tag{1}$$

which indicates that the perturbed value t' has a probability p of falling within the range $[t-d,t+d]$, and a probability of $\frac{p}{e^{\varepsilon}}$ of falling within the range $[d_1,t-d)\cup(t+d,d_2]$, where d_1 and d_2 are left and right boundaries of the domain.

According to probability theory, the expectation of t' is calculated as follows:

$$\begin{aligned} \mathrm{E}\left[t'\right] &= \int_{d_1}^{t-d}\frac{px}{e^{\varepsilon}}dx+\int_{t+d}^{d_2}\frac{px}{e^{\varepsilon}}dx+\int_{t-d}^{t+d}pxdx \\ &= \left(\frac{p}{2}-\frac{p}{2e^{\varepsilon}}\right)\cdot 4td+\frac{p}{2e^{\varepsilon}}\left({d_2}^2-{d_1}^2\right). \end{aligned}$$

Further, according to the equation for variance calculation, we can have

$$\begin{aligned} Var\left[t'\right] &= E\left[t'^2\right]-\left(E\left[t'\right]\right)^2 \\ &= \int_{d_1}^{t-d}\frac{px^2}{e^{\varepsilon}}dx+\int_{t+d}^{d_2}\frac{px^2}{e^{\varepsilon}}dx+\int_{t-d}^{t+d}px^2dx-\left(E\left[y_i\right]\right)^2 \\ &= p\left[\left(\frac{1}{3}-\frac{1}{3e^{\varepsilon}}\right)\left(6dt^2+2d^3\right)+\frac{1}{3e^{\varepsilon}}\left({d_2}^3-{d_1}^3\right)\right] \\ &\quad -\frac{p^2}{e^{2\varepsilon}}\begin{bmatrix}\left(e^{\varepsilon}-1\right)^2\cdot 4t^2d^2+\frac{1}{4}\left({d_2}^2-{d_1}^2\right)^2 \\ +\left(e^{\varepsilon}-1\right)\cdot 2td\cdot\left({d_2}^2-{d_1}^2\right)\end{bmatrix}. \end{aligned}$$

Therefore, we can obtain the following optimization problem:

$$\begin{aligned} &\underset{p,d}{\text{argmin}}\ \mathrm{Var}\left[t'\right] \\ &\text{s.t.}\quad 0<p<1 \\ &\qquad t-d\geqslant d_1 \\ &\qquad t+d\leqslant d_2 \\ &\qquad d>0. \end{aligned} \tag{2}$$

The probability p of the perturbation falling within the range $[t-d,t+d]$ and the value of d are determined by minimizing the variance. The constraints, respectively, bound the range of p and d, as well as the relationship between the left and right boundaries of the perturbation: $0<p<1$, $t-d\geq d_1$, $t+d\leq d_2$, and $d>0$. We can solve this using Fermat's theorem on stationary points, and we have

$$d=2pt^2\frac{\left(e^{\varepsilon}-1\right)}{e^{\varepsilon}}+\sqrt{4p^2t^4\frac{\left(e^{\varepsilon}-1\right)^2}{e^{2\varepsilon}}-t^2+\frac{pt}{e^{\varepsilon}}\left({d_2}^2-{d_1}^2\right)} \tag{3}$$

$$p=\frac{e^{\varepsilon}\cdot\left[\left(e^{\varepsilon}-1\right)\left(6dt^2+2d^3\right)+\left({d_2}^3-{d_1}^3\right)\right]}{6\begin{bmatrix}\left(e^{\varepsilon}-1\right)^2\cdot 4t^2d^2+\left(e^{\varepsilon}-1\right)\cdot 2td\cdot\left({d_2}^2-{d_1}^2\right) \\ +\frac{1}{4}\left({d_2}^2-{d_1}^2\right)^2\end{bmatrix}}. \tag{4}$$

C. Gaussian–Laplacian-Based MF

Current studies predominantly rely on Laplacian noise for privacy protection while overlooking the presence of Gaussian components within the noisy data, which can significantly impact model accuracy and robustness [16]. Since the noise generated by *BBV* shares great similarities with the Laplacian noise while meeting rigorous LDP, to address the above shortcoming, we propose a novel Gaussian–Laplacian mixture model-based MF method, termed *GLMF*, which combines both Laplacian and Gaussian noise models, resulting in a more comprehensive and

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nuanced treatment of data perturbations. Specifically, we have

$$P\left(R^*|u_i,v_j,\Pi,\Sigma,B,\alpha\right) = \prod_{i,j\in\Omega}\sum_{k=1}^{K}\pi_k\begin{bmatrix}\alpha N\left(r_{ij}^*|\left(u_i{}^T\right)v_j,{\sigma_k}^2\right)\\+(1-\alpha)\,L\left(r_{ij}^*|\left(u_i{}^T\right)v_j,b_k\right)\end{bmatrix}$$

where $P(R^*|u_i,v_j,\Pi,\Sigma,B,\alpha)$ denotes the probability of observing the perturbed rating matrix R^* given the user preference vector u_i, the item feature vector v_j, and the perturbed user–item rating matrix R^*. The hyperparameter α denotes the proportion of the Gaussian distribution in the mixture distribution.

$N(r_{ij}^*,(u_i)^Tv_j,\sigma_k^2)$ represents the pdf of the Gaussian distribution, indicating that the observed ratings r_{ij}^* follow a Gaussian distribution centered on the predicted ratings $(u_i)^Tv_j$ with a variance σ_k^2. Similarly, $L(r_{ij}^*,(u_i)^Tv_j,b_k)$ denotes the pdf of the Laplace distribution, indicating that the observed ratings r_{ij}^* follow a Laplace distribution centered on the predicted ratings $(u_i)^Tv_j$ with scale parameters b_k.

In addition, let $\Pi=(\pi_1,\pi_2,\ldots,\pi_K)$, $\Sigma=(\sigma_1,\sigma_2,\ldots,\sigma_K)$, and $B=(b_1,b_2,\ldots,b_K)$ indicate that r_{ij}^* $\gamma_{ijk}=1$ comes from the kth component, we have

$$\begin{aligned}&P\left(R^*|u_i,v_j,\Pi,\Sigma,B,\alpha\right)\\&=\prod_{k=1}^{K}\prod_{i,j\in\Omega}{\pi_k}^{\gamma_{ijk}}\begin{bmatrix}\alpha N\left(r_{ij}^*|\left(u_i{}^T\right)v_j,{\sigma_k}^2\right)\\+(1-\alpha)\,L\left(r_{ij}^*|\left(u_i{}^T\right)v_j,b_k\right)\end{bmatrix}^{\gamma_{ijk}}\\&=\prod_{k=1}^{K}{\pi_k}^{\beta_k}\prod_{i,j\in\Omega}\begin{bmatrix}\alpha\frac{1}{\sqrt{2\pi}\sigma_k}e^{-\frac{\left(r_{ij}^*-\left(u_i{}^T\right)v_j\right)^2}{2\sigma_k^2}}\\+(1-\alpha)\,\frac{1}{2b_k}e^{-\frac{\left|r_{ij}^*-\left(u_i{}^T\right)v_j\right|}{b_k}}\end{bmatrix}^{\gamma_{ijk}}.\end{aligned}$$

We define $\beta_k=\sum_{i,j\in\Omega}\gamma_{ijk}$ and $\sum_{i,j\in\Omega}\beta_k=|R^*|$. $|R^*|$ is the total number of nonmissing data points.

After maximizing the log posterior distribution over the eigenvector matrices of users and items by fixing the hyperparameters, we can obtain

$$\begin{aligned}&\ln P\left(R^*|u_i,v_j,\Pi,\Sigma,B,\alpha\right)\\&\geqslant\sum_{k=1}^{K}\left\{\begin{array}{l}\beta_k\ln\pi_k+\\\sum_{i,j\in\Omega}\gamma_{ijk}\end{array}\begin{bmatrix}\ln\left(\alpha\frac{1}{\sqrt{2\pi}\sigma_k}e^{-\frac{\left(r_{ij}^*-\left(u_i{}^T\right)v_j\right)^2}{2\sigma_k^2}}\right)\\+\ln\left((1-\alpha)\,\frac{1}{2b_k}e^{-\frac{\left|r_{ij}^*-\left(u_i{}^T\right)v_j\right|}{b_k}}\right)\end{bmatrix}\right\}.\end{aligned}$$

Based on the E-step of the EM, we have

$$\begin{aligned}\hat{\gamma}_{ijk}&=E_{\theta^{(q)}}\left[\gamma_{ijk}|{r_{ij}}^*,\theta\right]\\&=1*P\left(\gamma_{ijk}=1|r_{ij},\theta\right)+0*P\left(\gamma_{ijk}=0|r_{ij},\theta\right)\\&=\frac{\pi_k\begin{pmatrix}\alpha\frac{1}{\sigma_k\sqrt{2\pi}}e^{-\frac{\left(r_{ij}{}^*-\left(u_i{}^T\right)v_j\right)^2}{2{\sigma_k}^2}}\\+(1-\alpha)\,\frac{1}{2b_k}e^{-\frac{\left|r_{ij}{}^*-\left(u_i{}^T\right)v_j\right|}{b_k}}\end{pmatrix}}{\sum_{k=1}^{K}\pi_k\begin{pmatrix}\alpha\frac{1}{\sigma_k\sqrt{2\pi}}e^{-\frac{\left(r_{ij}{}^*-\left(u_i{}^T\right)v_j\right)^2}{2{\sigma_k}^2}}\\+(1-\alpha)\,\frac{1}{2b_k}e^{-\frac{\left|r_{ij}{}^*-\left(u_i{}^T\right)v_j\right|}{b_k}}\end{pmatrix}}.\end{aligned}$$

Then, based on the M-step of the EM, we have

$$\begin{cases}\hat{\pi}_k=\dfrac{\sum_{i.j\in\Omega}\hat{\gamma}_{ijk}}{|R^*|}\\\hat{\sigma}_k^2=\dfrac{\sum_{i.j\in\Omega}\hat{\gamma}_{ijk}\left(r_{ij}^*-\left(u_i{}^T\right)v_j\right)^2}{\sum_{i.j\in\Omega}\hat{\gamma}_{ijk}}\\\hat{b}_k=\dfrac{\sum_{i.j\in\Omega}\hat{\gamma}_{ijk}\left|r_{ij}^*-\left(u_i{}^T\right)v_j\right|}{\sum_{i.j\in\Omega}\hat{\gamma}_{ijk}}\\\hat{\alpha}=\frac{1}{2}.\end{cases}\tag{5}$$

Thus, we have

$$\begin{aligned}\arg\max_{U,V}&\sum_{k=1}^{K}\sum_{i,j\in\Omega}\gamma_{ijk}\begin{bmatrix}-\frac{\left(r_{ij}^*-\left(u_i{}^T\right)v_j\right)^2}{2\sigma_k^2}\\-\frac{\left|r_{ij}^*-\left(u_i{}^T\right)v_j\right|}{b_k}\end{bmatrix}\\&+\frac{\lambda}{2}\sum_{i=1}^{N}\left\|U_i\right\|_{Fro}^2+\frac{\lambda}{2}\sum_{j=1}^{M}\left\|V_j\right\|_{Fro}^2.\end{aligned}\tag{6}$$

Algorithm 1 presents the details of *MENTOR*.

Algorithm 1: MENTOR.

Input: ε: total privacy budget; R: non-private ratings; K: number of mixed distributions; λ: regularization parameter;
Output: U and V;
Upload noisy ratings;
Randomly initialize U, V, Π ,Σ, B and α;
E-step: Estimate posterior responsibility γ_{ijk} using the current model parameters;
while *Until Convergence* **do**
  M-Step: Update parameters Π ,Σ, B and α;
  M-Step: Update latent factor matrices U and V;
  E-step: Recalculate posterior responsibility γ_{ijk};
return: Updated latent factor matrices U and V;

V. Algorithm Analysis

According to (1), *MENTOR* guarantees rigorous LDP, as evidenced. Notably, *MENTOR* achieves this privacy guarantee without incurring additional communication overhead compared to the baseline *PMF* method, which is the same as [15].

Theorem 1: Let U_{Lap} and V_{Lap} represent the user and item latent spaces considering only Laplacian noise, respectively. We have established that $|U_{\text{MENTOR}}-U_{\text{PMF}}|<|U_{\text{Lap}}-U_{\text{PMF}}|$ and $|V_{\text{MENTOR}}-V_{\text{PMF}}|<|V_{\text{Lap}}-V_{\text{PMF}}|$, which indicates that the latent space derived by *MENTOR* exhibits greater proximity

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to the nonprivate weight U_{PMF} than the Laplacian noise models alone.

Proof: The iteration formulas for U_{PMF} and U_{Lap} are as follows:

$$U_{\text{PMF}} \leftarrow U_i + \eta \left[\sum_{j=1}^{M} \left(R_{ij} - U_i^T V_j \right) V_j - \lambda U_i \right] \tag{7}$$

$$U_{\text{Lap}} \leftarrow U_i + \eta \left[\sum_{j=1}^{M} \left[\begin{matrix} \left(R_{ij}^* - U_i^T V_j \right) \\ + \frac{1}{R_{ij}^* - U_i^T V_j} \end{matrix} \right] V_j - \lambda U_i \right]. \tag{8}$$

We can obtain

$$\begin{aligned} &U_{\text{MENTOR}} - U_{\text{PMF}} \\ &= \eta \textstyle\sum_{k=1}^{K} \sum_{j=1}^{M} \left[\begin{matrix} \frac{\left(R_{ij}^* - U_i^T V_j \right)}{\sigma_k^2} + \frac{\text{sgn}\left(R_{ij}^* - U_i^T V_j \right)}{b_k} \\ -K \left(R_{ij} - U_i^T V_j \right) \end{matrix} \right] V_j \end{aligned} \tag{9}$$

$$U_{\text{Lap}} - U_{\text{PMF}} = \eta \sum_{j=1}^{M} \left[\left(R_{ij}^* - R_{ij} \right) + \frac{1}{R_{ij}^* - U_i^T V_j} \right] V_j. \tag{10}$$

1) When $R_{ij}^* > U_i^T V_j$: According to (9) and (10), it is known that $U_{\text{MENTOR}} - U_{\text{PMF}} < 0$ and $U_{\text{Lap}} - U_{\text{PMF}} > 0$, then

$$\begin{aligned} &\left| U_{\text{MENTOR}} - U_{\text{PMF}} \right| - \left| U_{\text{Lap}} - U_{\text{PMF}} \right| \\ &= \left(U_{\text{PMF}} - U_{\text{MENTOR}} \right) - \left(U_{\text{Lap}} - U_{\text{PMF}} \right) \\ &= 2U_{\text{PMF}} - U_{\text{MENTOR}} - U_{\text{Lap}} \\ &= \eta \sum_{k=1}^{K} \sum_{j=1}^{M} \left[\begin{matrix} K \left(R_{ij} - U_i^T V_j \right) - K \left(R_{ij}^* - R_{ij} \right) \\ -\frac{\left(R_{ij}^* - U_i^T V_j \right)}{\sigma_k^2} - \frac{1}{b_k} - \frac{K}{R_{ij}^* - U_i^T V_j} \end{matrix} \right] V_j \\ &< 0. \end{aligned} \tag{11}$$

2) When $R_{ij}^* < U_i^T V_j$: According to (9) and (10), it is known that $U_{\text{MENTOR}} - U_{\text{PMF}} < 0$ and $U_{\text{Lap}} - U_{\text{PMF}} < 0$, then

$$\begin{aligned} &\left| U_{\text{MENTOR}} - U_{\text{PMF}} \right| - \left| U_{\text{Lap}} - U_{\text{PMF}} \right| \\ &= \left(U_{\text{PMF}} - U_{\text{MENTOR}} \right) - \left(U_{\text{PMF}} - U_{\text{Lap}} \right) \\ &= U_{\text{Lap}} - U_{\text{MENTOR}} \\ &= \eta \sum_{k=1}^{K} \sum_{j=1}^{M} \left[\begin{matrix} K \left(R_{ij}^* - U_i^T V_j \right) + \frac{K}{R_{ij}^* - U_i^T V_j} \\ -\frac{\left(R_{ij}^* - U_i^T V_j \right)}{\sigma_k^2} + \frac{1}{b_k} \end{matrix} \right] V_j < 0. \end{aligned} \tag{12}$$

It follows the same way we have $|V_{\text{MENTOR}} - V_{\text{PMF}}| < |V_{\text{Lap}} - V_{\text{PMF}}|$.

VI. Experiments

A. Experimental Setup

Datasets: We conduct comprehensive experiments on the following two real-world datasets.

1) *Epinions:*[1] It, *Epi* for short, consists of 49 289 users, 139 738 items, and 664 824 rating records.
2) *Ciao:*[2] It, *Epi* for short, consists of 7375 users, 106 797 items, and 284 086 rating records.

Evaluation metrics and baselines: For utility measure, we adopt the commonly used *RMSE* and *F-Measure*. We also use *RMSE* to measure the approximation between the nonprivate data and the Laplace distribution to verify the rationality of the Laplace distribution. For the baselines, we make a comparison with the following state-of-the-art competitors.

1) *NoPriv* [19]: The objective function is the same to *PMF*.
2) *PWMF* [16]: It adopts *Piecewise* [22] to protect the gradients and thus to protect the ratings.
3) *PGMM* [15]: It incorporates GMM to design a new MF algorithm. As suggested by their paper, the number of Gaussian components is set to 5.
4) *GCNMF* [17]: It uses a graph convolutional network to update model parameters to achieve personalized LDP.

Implementation details: The range of values for ε is $[0.1, 1.0]$. The number of mixed distributions is set to $K = 5$. For a fair comparison, as a common sense, the regularization parameter for all competitors is set to $\lambda = 0.001$, and the learning rate for *SGD* is also set to 0.001. Moreover, we set the stop threshold of *EM* as 0.0001. The dimension for latent space is set to 15. In total, 80% will be used as the training set, and the remaining will be used as the testing set, and the average values are reported.

B. Performance Comparison

Impact of ε: Fig. 2(a) and (b) shows, respectively, the performance with different privacy budgets ε by *RMSE* and *F-Measure* in *Epi*. Fig. 2(c) and (d) shows for Ciao. Recall that *NoPriv* denotes the nonprivate algorithm *PMF*. We summarize our findings and discuss the reasons.

1) Not surprisingly, the performance of all the approaches becomes better when enlarging the privacy budget ε, as the scale of injected noise is inversely proportional to the size of ε, and larger ε results in smaller noise injection.
2) *MENTOR* consistently significantly outperforms the competitors on both datasets in all cases and is most closely approximated to that of the *NoPriv*. Specifically, *MENTOR* outperforms the state of the art by at least 15% and 12% improvement with regard to *RMSE* and *F-Measure*. The reason can be explained as follows. The scale of injected noise is directly proportional to the sensitivity of the data, with smaller sensitivity resulting in smaller noise injection. We limit the sensitivity of the data through *BBV* while minimizing the variance to reduce the overall injection noise. Moreover, integrating mixed noise into traditional MF will make *MENTOR* more robust against noise.
3) For *PWMF*, it may suffer from unstable prediction accuracy due to the perturbation of gradients. In addition, users compute their latent space locally without submitting

[1][Online]. Available: https://snap.stanford.edu/data/soc-Epinions1.html
[2][Online]. Available: https://www.kaggle.com/datasets/aravindaraman/ciao-data/data

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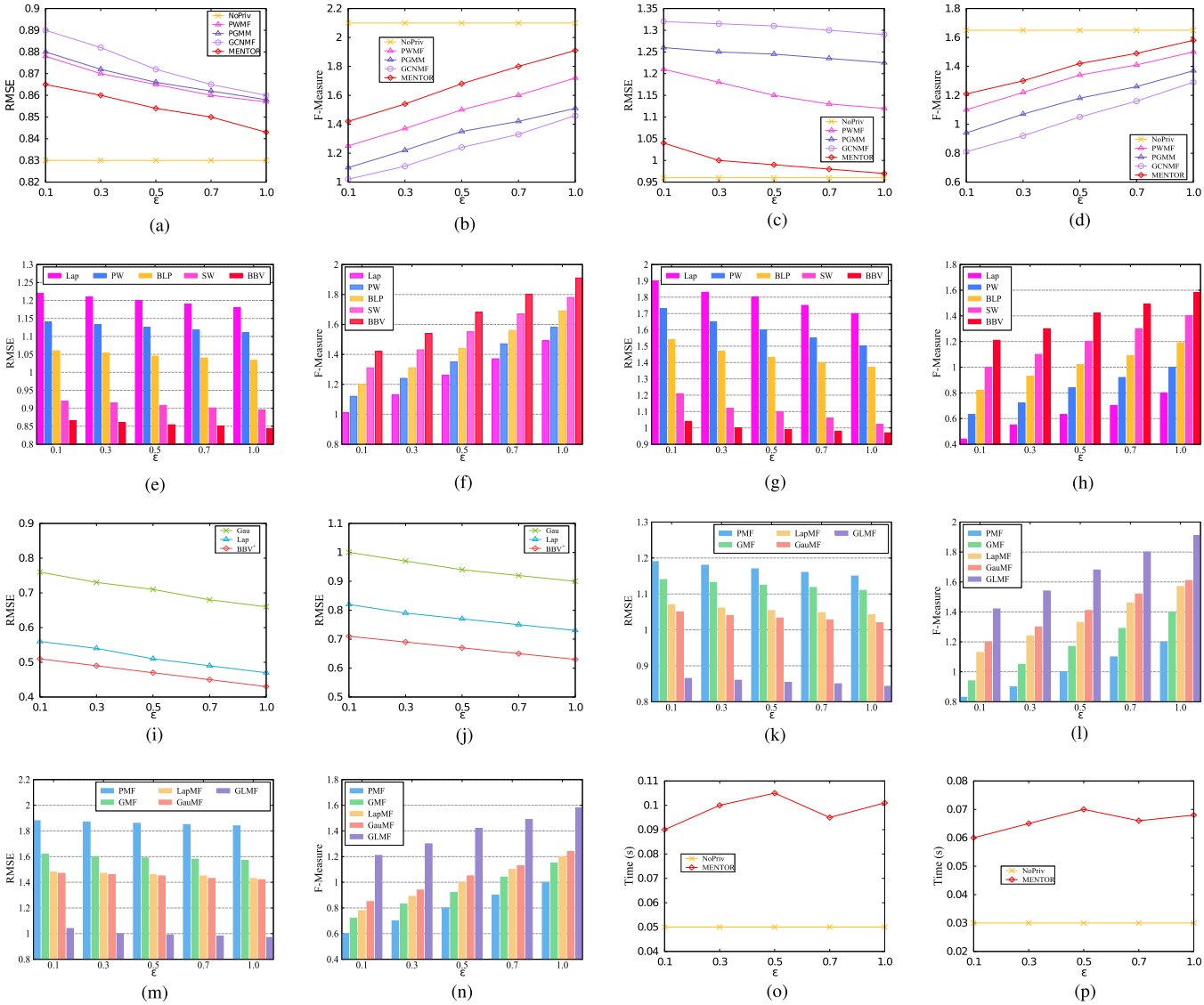


Fig. 2. Performance comparison. (a) *MENTOR: RMSE* on *Epi*. (b) *MENTOR: F-Measure* on *Epi*. (c) *MENTOR: RMSE* on *Ciao*. (d) *MENTOR: F-Measure* on *Ciao*. (e) *BBV: RMSE* on *Epi*. (f) *BBV: F-Measure* on *Epi*. (g) *BBV: RMSE* on *Ciao*. (h) *BBV: F-Measure* on *Ciao*. (i) BBV^{+}: *RMSE* on *Epi*. (j) BBV^{+}: *RMSE* on *Ciao*. (k) *GLMF: RMSE* on *Epi*. (l) *GLMF: F-Measure* on *Epi*. (m) *GLMF: RMSE* on *Ciao*. (n) *GLMF: F-Measure* on *Ciao*. (o) *MENTOR:* Time on *Epi*. (p) *MENTOR:* Time on *Ciao*.

it and then update the item's latent space after collecting gradients under LDP. This process involves iterative communication between users and the server, resulting in significant communication overhead. For *PGMM*, it employs a customized approach that only models the Gaussian component for the uploaded noisy data, neglecting other noise types, which limits the practical deployment of this method. For *GCNMF*, it may only be effective for graph-structured data and often involves a clustering algorithm. Since the number of clusters cannot be determined in advance, the recommendation accuracy might be unstable.

4) Note that *GCNMF* guarantees personalized LDP, which may be treated as a weaker version of LDP. The reason is that different users have different privacy budgets, but they often declare that they guarantee the strength of privacy protection under the minimum privacy budget. Therefore, under the same privacy budget, most users may have a larger privacy budget. However, even with these privacy relaxations, we observe that *MENTOR* still significantly outperforms it.

Effectiveness of BBV: Fig. 2(e)–(h) illustrates the effectiveness of *BBV*. *Lap*, *PW*, *BLP*, and *SW* represent Laplacian noise, *Piecewise* [22], bounded Laplacian noise [28], and square wave [29]. We have the following observations. First, all the methods outperform *Lap*, which validates the necessity of introducing bounded noise. Second, although *BLP* is also bounded noise, it does not minimize injection noise. That is, it does not minimize variance. *SW* limits the true value to its vicinity with a high probability, so the perturbed value may not be far from the true value,

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resulting in better performance. At last, *BBV* always performs best when ε gradually enlarges. This is because of its ability to add noise in a controlled manner within predefined bounds in both directions, minimizing the impact of extreme noise values. Since *BBV* serving as the input perturbation mechanism does play a leading role, it can retain data utility more effectively while still ensuring strong privacy guarantees.

Rationality of Laplace distribution: To verify the rationality of the Laplace distribution, we conduct experiments in this section and verify the difference between data with different noise levels and nonprivate data, where *Gau* represents the Gaussian mechanism and the probability violating LDP is set to 0.0001. BBV^+ indicates that, compared to *BBV*, we only report the difference between raw data and noise data, rather than prediction accuracy. Fig. 2(i) and (j) shows the results. It can be seen that the data after *BBV* are very close to the Laplace distribution, which is consistent with general intuition. That is, the Laplace distribution satisfies the LDP.

Effectiveness of GLMF: Fig. 2(k)–(n) illustrates the effectiveness of *GLMF*. We compare four MF methods for rating prediction: *PMF* (noise-free modeling), *LapMF* (only Laplacian noise), *GMF* (Gaussian distribution assumption for initializing latent spaces), and *GauMF* (only Gaussian noise introduction). Our observations are as follows. First, all methods outperform *PMF*, demonstrating the importance of noise modeling. Second, *GauMF* surpasses other methods by introducing Gaussian noise into the factorization process, thereby enhancing the model's robustness and accuracy in dealing with errors and uncertainties via *Euclidean distance*, resulting in more stable and reliable recommendations compared to the *absolute distance* used by *LapMF*. Finally, *GLMF* outperforms all other predictive models by considering not only the injected noise but also the inherent Gaussian noise in the data, which represents user errors and leads to more realistic modeling.

Efficiency study: We implemented all the approaches in Python on an Intel i5-8250 U 2.60 GHz *Laptop*. Fig. 2(o) and (p) shows the results. Since the running time on *Epi* is over 2 h, we just report the time for each prediction. We observe that the running time of *MENTOR* is slightly larger than *NoPriv*, indicating that the privacy protection measures in *MENTOR* only slightly increase the runtime overhead and are thus suitable for large-scale deployment.

VII. Conclusion

This article presents *MENTOR* that addresses fundamental challenges in MF under LDP. Our work tackles three major problems: accuracy degradation from noise injection, communication overhead from multiple user interactions, and the compound effect of both injected and inherent noise in recommendation systems. We introduce two key innovations: *BBV*, a bidirectional bounded input perturbation mechanism that minimizes noise variance while maintaining LDP guarantees, and *GLMF*, a sophisticated MF approach based on Gaussian–Laplacian mixture modeling to handle various types of data uncertainties.

For future work, we identify two promising directions: 1) adapting *MENTOR* for online learning scenarios to enable real-time updates to recommendations while maintaining privacy guarantees, and 2) investigating the incorporation of side information, such as user or item features, into the *MENTOR* framework without compromising privacy, potentially further enhancing recommendation accuracy. These extensions could significantly broaden the applicability and effectiveness of *MENTOR* in real-world, dynamic recommendation environments.

References

[1] S. Meng et al., "Privacy-aware factorization-based hybrid recommendation method for healthcare services," *IEEE Trans. Ind. Inform.*, vol. 18, no. 8, pp. 5637–5647, Aug. 2022.

[2] C. Wei, C. Hu, C.-D. Wang, and S. Huang, "Time-aware multibehavior contrastive learning for social recommendation," *IEEE Trans. Ind. Inform.*, vol. 20, no. 4, pp. 6424–6435, Apr. 2024.

[3] J. Feng, Y. Wu, H. Sun, S. Zhang, and D. Liu, "Panther: Practical secure two-party neural network inference," *IEEE Trans. Inf. Forensics Secur.*, vol. 20, pp. 1149–1162, 2025.

[4] G. Li, Z. Cai, and G. Yin, "Differentially private recommendation system based on community detection in social network applications," *Secur. Commun. Netw.*, vol. 2018, no. 1, 2018, Art. no. 3530123.

[5] A. Friedman, S. Berkovsky, and M. Ali Kâafar, "A differential privacy framework for matrix factorization recommender systems," *User Model. User Adapted Interact.*, vol. 26, no. 5, pp. 425–458, 2016.

[6] S. Gao, X. Di Ma, J. Zhu, and J. Ma, "APRS: A privacy-preserving location-aware recommender system based on differentially private histogram," *Sci. China Inf. Sci.*, vol. 60, no. 11, pp. 119103:1–119103:3, 2017.

[7] S. Yang, K. Zhu, and W. Liang, "Differential privacy for context-aware recommender systems," in *Proc. IEEE 18th Int. Conf. Cogn. Inform. Cogn. Comput.*, 2019, pp. 356–360.

[8] J. Ren, X. Xu, Z. Yao, and H. Yu, "Recommender systems based on autoencoder and differential privacy," in *Proc. IEEE 43rd Annu. Comput. Softw. Appl. Conf.*, 2019, pp. 358–363.

[9] S. Yang and K. Zhu, "Differential privacy and Bayesian for context-aware recommender systems," *Int. J. Cogn. Inform. Nat. Intell.*, vol. 15, no. 4, pp. 1–13, 2021.

[10] A. G. Carranza, R. Farahani, and N. Ponomareva, "Privacy-preserving recommender systems with synthetic query generation using differentially private large language models," 2023, *arXiv:2305.05973*.

[11] P. Müllner, E. Lex, and M. Schedl, "Differential privacy in collaborative filtering recommender systems: A review," *Front. Big Data*, vol. 6, 2024, Art. no. 1249997.

[12] W. Li, H. Chen, R. Zhao, and C. Hu, "A federated recommendation system based on local differential privacy clustering," in *Proc. IEEE SmartWorld, Ubiquitous Intell. Comput., Adv. Trusted Comput., Scalable Comput., Commun., Int. People Smart City Innov.*, 2021, pp. 364–369.

[13] A. Liu, Y. Yao, and X. Cheng, "Recommender systems with condensed local differential privacy," in *Proc. Int. Conf. Mach. Learn. Cyber Secur.*, 2020, pp. 355–365.

[14] X. Liu, A. Liu, and X. Zhang, "When differential privacy meets randomized perturbation: A hybrid approach for privacy-preserving recommender system," in *Proc. Database Syst. Adv. Appl.: 22nd Int. Conf.*, 2017, pp. 576–591.

[15] J. Neera, X. Chen, N. Aslam, K. Wang, and Z. Shu, "Private and utility enhanced recommendations with local differential privacy and Gaussian mixture model," *IEEE Trans. Knowl. Data Engine.*, vol. 35, no. 4, pp. 4151–4163, Apr. 2023.

[16] Y. Wang, M. Gao, and X. Ran, "An improved matrix factorization with local differential privacy based on piecewise mechanism for recommendation systems," *Expert Syst. Appl.*, vol. 216, 2023, Art. no. 119457.

[17] C. Chen, G. Zhang, B. Liu, and W. Dou, "A GCN-based model for recommendation using local differential privacy method," in *Proc. IEEE Int. Conf. High Perform. Comput., Commun., Data Sci., Syst., Smart City, Dependability Sensor, Cloud, Big Data Syst., Appl.*, 2023, pp. 194–201.

[18] H. Liu et al., "EDMF: Efficient deep matrix factorization with review feature learning for industrial recommender system," *IEEE Trans. Ind. Inform.*, vol. 18, no. 7, pp. 4361–4371, Jul. 2022.

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[19] J. Strahl, J. Peltonen, and H. Mamitsuka, "Scalable probabilistic matrix factorization with graph-based priors," in *Proc. AAAI conf. Artif. Intell.*, 2020, pp. 5851–5858.
[20] C. Gao, C. Huang, and D. Lin, "DPLCF: Differentially private local collaborative filtering," in *Proc. 43rd Int. ACM SIGIR Conf. Res. Develop. Inf. Retrieval*, 2020, pp. 961–970.
[21] X. Zheng, M. Guan, X. Jia, L. Guo, and Y. Luo, "A matrix factorization recommendation system-based local differential privacy for protecting users' sensitive data," *IEEE Trans. Comput. Social Syst.*, vol. 10, no. 3, pp. 1189–1198, Jun. 2023.
[22] N. Wang et al., "Collecting and analyzing multidimensional data with local differential privacy," in *Proc. IEEE 35th Int. Conf. Data Eng.*, 2019, pp. 638–649.
[23] H. Zhou, G. Yang, Y. Xiang, Y. Bai, and W. Wang, "A lightweight matrix factorization for recommendation with local differential privacy in Big Data," *IEEE Trans. Big Data*, vol. 9, no. 1, pp. 160–173, Feb. 2023.
[24] X. Meng, S. Wang, and K. Shu, "Towards privacy preserving social recommendation under personalized privacy settings," *World Wide Web*, vol. 22, pp. 2853–2881, 2019.
[25] Y. Huo, B. Chen, and T. Jing, "Privacy-preserving point-of-interest recommendation based on geographical and social influence," *Inform. Sci.*, vol. 543, pp. 202–218, 2021.
[26] W. Wu, X. Yang, and R. Lei, "A trust-delivery collaborative recommendation method based on local differential privacy," in *Proc. 26th Int. Conf. Comput. Supported Cooperative Work Des.*, 2023, pp. 1074–1079.
[27] C. Xu, X. Mei, and D. Liu, "An efficient privacy-preserving point-of-interest recommendation model based on local differential privacy," *Complex Intell. Syst.*, vol. 9, no. 3, pp. 3277–3300, 2023.
[28] N. Holohan, S. Antonatos, and S. Braghin, "The bounded Laplace mechanism in differential privacy," *J. Priv. Confidentiality*, vol. 10, no. 1, 2020.
[29] Z. Li, T. Wang, and M. Lopuha, "Estimating numerical distributions under local differential privacy," in *Proc. ACM SIGMOD Int. Conf. Manage. Data*, 2020, pp. 621–635.

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